Table 3
Tensile properties of interply hybrid composites for experiments and predictions.

Code	Stacking sequence	Geometrical properties			Modulus		Strength		Failure strain
		t_c (mm)	V_f (%)	V_v (%)	Experiment (GPa)	Expected (GPa)	Experiment (MPa)	Expected (MPa)	Experiment (%)
CFRPP 6	$[CFRPP]_3$	0.89	55	3	50.7	–	644	–	1.19
SRPP 1	$[SRPP]_6$	0.91	0	–	2.8	–	149	–	21.7
Hybrid 1	$[SRPP_2/CFRPP/SRPP_2]$	0.96	17	3	15.4 (18%↓)	19	219 (5%↓)	231	1.27 (7%↑)
Hybrid 2	$[CFRPP/SRPP]_s$	0.99	33	5	26.3 (24%↓)	35	398 (9%↓)	438	1.41 (18%↑)
Hybrid 3	[SRPP/CFRPP/SRPP/CFRPP/SRPP]	1.14	28	4	22.8 (25%↓)	30	347 (8%↓)	378	1.31 (10%↑)

Table 4
Comparison of theoretical compressive pre-strain and improved failure strain among various interply hybrid composites.

Code	Stacking sequence	Volume fraction of SRPP (%)	Theoretical pre-strain (%)	Improved failure strain (%)
Hybrid 1	$[SRPP_2/CFRPP/SRPP_2]$	66	0.21	0.08
Hybrid 2	$[CFRPP/SRPP]_s$	33	0.11	0.22
Hybrid 3	[SRPP/CFRPP/SRPP/CFRPP/SRPP]	43	0.13	0.12

strength. Since the stress–strain curves are almost linear, the hybrid composites could improve the material potential in terms of strain energy.

5. Conclusions

In this research, interply hybrid composites with carbon fiber reinforced polypropylene (CFRPP) and self-reinforced polypropylene (SRPP) were investigated. SRPP has an intrinsic behavior of shrinkage under high temperature, and hence could induce a compressive pre-strain in the carbon fiber containing CFRPP layers during consolidation of the CFRPP/SRPP hybrid composite. Therefore, we expected a positive hybrid effect on the tensile failure, which is anyway dominated by the failure strain of the carbon fibers.

First, CFRPP was prepared by PP film stacking impregnation of carbon fabric. The influence of the processing conditions on the impregnation quality and the tensile properties was investigated, optimizing the conditions with respect to maximizing tensile sustainable stress per carbon fiber volume. Second, SRPP produced by hot compaction of highly oriented PP woven fabric was characterized. The influence of processing conditions on tensile properties and recovery stress was investigated. Finally, the interply hybrid composites with CFRPP and SRPP were produced under the optimized condition. Tensile testing of the composites revealed that the failure strain improvement was in reasonable agreement with the compressive pre-strain induced by SRPP shrinkage. The modulus and strength, however, were found to be lower than the ones predicted from the rule of mixture. Local carbon fiber misalignment caused by the SRPP shrinkage during the consolidation may reduce the stiffness and hence to hinder the expected hybrid effect. However, in terms of failure strain, a positive hybrid effect of +7% to +18% could be obtained. Further investigations are being carried out to evaluate the effect of the compressive pre-strain on other damage tolerance characterization like the strain to matrix crack initiation in tensile and fatigue loading, and the impact resistance. Results of these studies will be reported in a next paper.

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